

An error model for odometric position estimation

Module 4

An error model for odometric position estimation

Generally the pose (position) of a robot is represented by the vector

$$p = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \quad \text{-----} \quad (1)$$

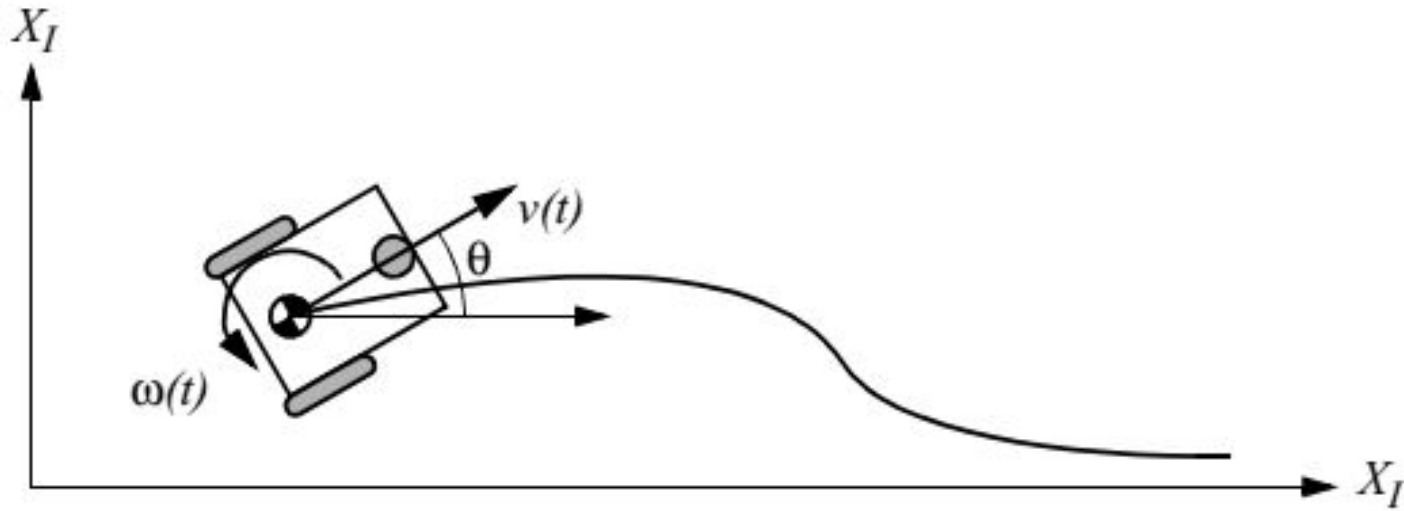


Figure 5.3
Movement of a differential-drive robot.

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- For a differential-drive robot the position can be estimated starting from a known position by integrating the movement (summing the incremental travel distances).
- For a discrete system with a fixed sampling interval the incremental travel distances $(\Delta x; \Delta y; \Delta \theta)$ are

$$\Delta x = \Delta s \cos(\theta + \Delta \theta / 2) \quad \text{————— (2)}$$

$$\Delta y = \Delta s \sin(\theta + \Delta \theta / 2) \quad \text{————— (3)}$$

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$$\Delta\theta = \frac{\Delta s_r - \Delta s_l}{b} \quad (4)$$

$$\Delta s = \frac{\Delta s_r + \Delta s_l}{2} \quad (5)$$

where

$(\Delta x; \Delta y; \Delta\theta)$ = path traveled in the last sampling interval;

$\Delta s_r; \Delta s_l$ = traveled distances for the right and left wheel respectively;

b = distance between the two wheels of differential-drive robot.

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- Thus we get the updated position p'

$$p' = \begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = p + \begin{bmatrix} \Delta s \cos(\theta + \Delta\theta/2) \\ \Delta s \sin(\theta + \Delta\theta/2) \\ \Delta\theta \end{bmatrix} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \Delta s \cos(\theta + \Delta\theta/2) \\ \Delta s \sin(\theta + \Delta\theta/2) \\ \Delta\theta \end{bmatrix} \quad (6)$$

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- By using the relation for $(\Delta s; \Delta \theta)$ of equations (5) and (4) we further obtain the basic equation for odometric position update (for differential drive robots):

$$p' = f(x, y, \theta, \Delta s_r, \Delta s_l) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{\Delta s_r + \Delta s_l}{2} \cos\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r + \Delta s_l}{2} \sin\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r - \Delta s_l}{b} \end{bmatrix} \quad (7)$$

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- odometric position updates can give only a very rough estimate of the actual position.
- Owing to integration errors of the uncertainties of P and the motion errors during the incremental motion $(\Delta s_r; \Delta s_l)$ the position error based on odometry integration grows with time.

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- In the next step we will establish an error model for the integrated position p' to obtain the covariance matrix $\Sigma_{p'}$ of the odometric position estimate.
- To do so, we assume that at the starting point the initial covariance matrix Σ_p is known.
- For the motion increment $(\Delta s_r; \Delta s_l)$ we assume the following covariance matrix Σ_{Δ}

Covariance matrix

- In probability theory and statistics, a covariance matrix is a square matrix giving the covariance between each pair of elements of a given random vector.
- the covariance matrix generalizes the notion of variance to multiple dimensions.

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$$\Sigma_{\Delta} = covar(\Delta s_r, \Delta s_l) = \begin{bmatrix} k_r |\Delta s_r| & 0 \\ 0 & k_l |\Delta s_l| \end{bmatrix} \quad (8)$$

- where Δs_r and Δs_l are the distances traveled by each wheel, and k_r , k_l are error constants representing the nondeterministic parameters of the motor drive and the wheel-floor interaction.

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- In equation 8 we made the following assumptions:
 1. The two errors of the individually driven wheels are independent
 2. The variance of the errors (left and right wheels) are proportional to the absolute value of the traveled distances (Δs_r ; Δs_l)

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- These assumptions, while not perfect, but these are suitable and will thus be used for the further development of the error model.
- The motion errors are due to imprecise movement because of deformation of wheel, slippage, unequal floor, errors in encoders, and so on.
- The values for the error constants k_r and k_l depend on the robot and the environment and should be experimentally established by performing and analyzing representative movements.

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- If we assume that p and $\Delta_{rl} = (\Delta s_r ; \Delta s_l)$ are uncorrelated and the derivation of f [equation (7)] is reasonably approximated by the first-order Taylor expansion (linearization), we conclude, using the error propagation law

$$\Sigma_{p'} = \nabla_p f \cdot \Sigma_p \cdot \nabla_p f^T + \nabla_{\Delta_{rl}} f \cdot \Sigma_{\Delta} \cdot \nabla_{\Delta_{rl}} f^T$$

gradient / divergence operator-del

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- The covariance matrix Σ_p is, of course, always given by the Σ_p of the previous step, and can thus be calculated after specifying an initial value .
- Using equation (7) we can develop the two Jacobians

$$F_p = \nabla_p f \text{ and } F_{\Delta_{rl}} = \nabla_{\Delta_{rl}} f :$$

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- Thus we get the updated position p'

$$p' = \begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = p + \begin{bmatrix} \Delta s \cos(\theta + \Delta\theta/2) \\ \Delta s \sin(\theta + \Delta\theta/2) \\ \Delta\theta \end{bmatrix} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \Delta s \cos(\theta + \Delta\theta/2) \\ \Delta s \sin(\theta + \Delta\theta/2) \\ \Delta\theta \end{bmatrix} \quad (6)$$

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$$F_p = \nabla_p f = \nabla_p (f^T) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial \theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\Delta s \sin(\theta + \Delta\theta/2) \\ 0 & 1 & \Delta s \cos(\theta + \Delta\theta/2) \\ 0 & 0 & 1 \end{bmatrix}$$

(10)

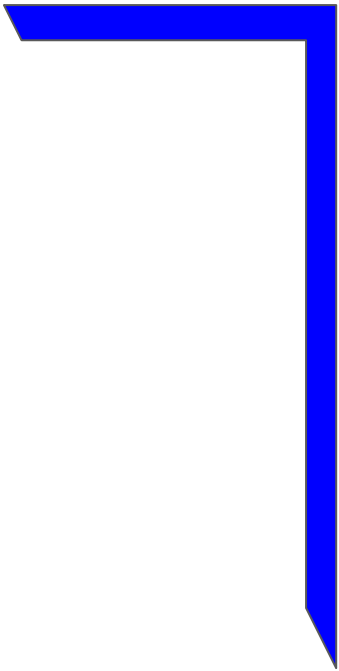
$$p' = f(x, y, \theta, \Delta s_r, \Delta s_l) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{\Delta s_r + \Delta s_l}{2} \cos\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r + \Delta s_l}{2} \sin\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r - \Delta s_l}{b} \end{bmatrix} \quad (7)$$

$$F_{\Delta_{rl}} = \begin{bmatrix} \frac{1}{2} \cos\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta s}{2b} \sin\left(\theta + \frac{\Delta\theta}{2}\right) & \frac{1}{2} \cos\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta s}{2b} \sin\left(\theta + \frac{\Delta\theta}{2}\right) \\ \frac{1}{2} \sin\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta s}{2b} \cos\left(\theta + \frac{\Delta\theta}{2}\right) & \frac{1}{2} \sin\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta s}{2b} \cos\left(\theta + \frac{\Delta\theta}{2}\right) \\ \frac{1}{b} & -\frac{1}{b} \end{bmatrix} \quad (11)$$

The details for arriving at equation (11) are

$$F_{\Delta_{rl}} = \nabla_{\Delta_{rl}} f = \left[\frac{\partial f}{\partial \Delta s_r} \quad \frac{\partial f}{\partial \Delta s_l} \right] = \dots \quad (12)$$

$$\begin{bmatrix} \frac{\partial \Delta s}{\partial \Delta s_r} \cos\left(\theta + \frac{\Delta \theta}{2}\right) + \frac{\Delta s}{2} \sin\left(\theta + \frac{\Delta \theta}{2}\right) \frac{\partial \Delta \theta}{\partial \Delta s_r} & \frac{\partial \Delta s}{\partial \Delta s_l} \cos\left(\theta + \frac{\Delta \theta}{2}\right) + \frac{\Delta s}{2} \sin\left(\theta + \frac{\Delta \theta}{2}\right) \frac{\partial \Delta \theta}{\partial \Delta s_l} \\ \frac{\partial \Delta s}{\partial \Delta s_r} \sin\left(\theta + \frac{\Delta \theta}{2}\right) + \frac{\Delta s}{2} \cos\left(\theta + \frac{\Delta \theta}{2}\right) \frac{\partial \Delta \theta}{\partial \Delta s_r} & \frac{\partial \Delta s}{\partial \Delta s_l} \sin\left(\theta + \frac{\Delta \theta}{2}\right) + \frac{\Delta s}{2} \cos\left(\theta + \frac{\Delta \theta}{2}\right) \frac{\partial \Delta \theta}{\partial \Delta s_l} \\ \frac{\partial \Delta \theta}{\partial \Delta s_r} & \frac{\partial \Delta \theta}{\partial \Delta s_l} \end{bmatrix} \quad (13)$$



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